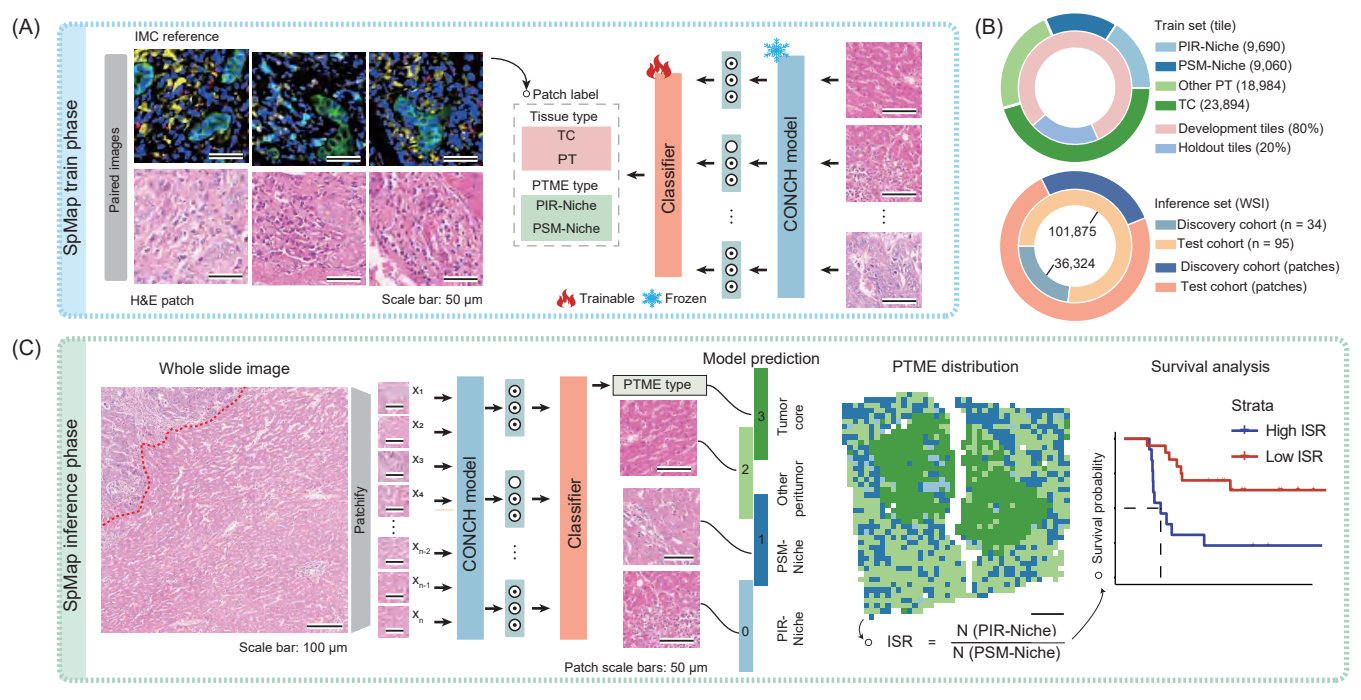

(A)
SpMap train phase
IMC reference
Paired images
H&E patch
Scale bar: 50 μm
Patch label
Tissue type
TC
PT
PTME type
PIR-Niche
PSM-Niche
Classifier
CONCH model
Trainable
Frozen
(B)
Train set (tile)
PIR-Niche (9,690)
PSM-Niche (9,060)
Other PT (18,984)
TC (23,894)
Development tiles (80%)
Holdout tiles (20%)
Inference set (WSI)
101,875
36,324
Discovery cohort (n = 34)
Test cohort (n = 95)
Discovery cohort (patches)
Test cohort (patches)
(C)
SpMap inference phase
Whole slide image
Scale bar: 100 μm
Patchify
CONCH model
Classifier
PTME type
Model prediction
3
2
1
0
Tumor core
Other peritumor
PSM-Niche
PIR-Niche
Patch scale bars: 50 μm
PTME distribution
ISR = N (PIR-Niche) / N (PSM-Niche)
Survival analysis
Strata
High ISR
Low ISR
Survival probability
0